

Job: _____

Engineer: _____

Contractor: _____

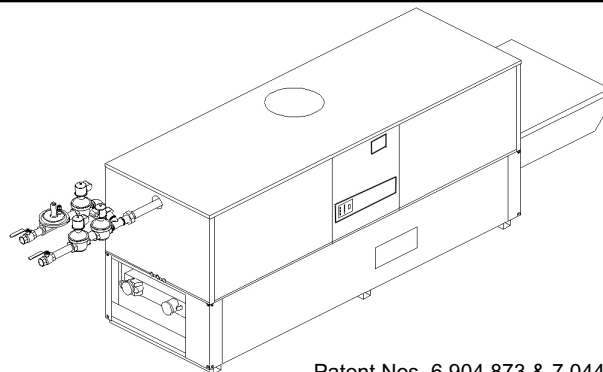
Prepared By: _____ Date: _____

Model: _____ Indoor/Outdoor: _____ MBTUH Input: _____

Hi Delta FlexGas® - Type WH

Dual Fuel Water Heaters Models 302CD-902CD

- ♦ 100% Factory Fire Tested
- ♦ Thermal Efficiency: Up to 85%
- ♦ Versa IC® Controller with LCD Display
- ♦ Maximum Setpoint: 150°F
- ♦ Maximum Outlet Temperature: 200°F
- ♦ Minimum Non-Condensing Inlet Temperature: 105°F
- ♦ Thermal Shock Proof Heat Exchanger
- ♦ Limited Twenty-Year Thermal Shock Warranty
- ♦ Limited Ten-Year Cupro-Nickel Heat Exchanger Warranty
- ♦ Limited Five-Year Copper Heat Exchanger Warranty
- ♦ Full Safety Diagnostics with Fault History
- ♦ PolyTuf Powder Coated Cabinet
- ♦ No Combustible Floor Shield Required
- ♦ Fan-Assisted
- ♦ Patented Burner Security Blanket
- ♦ Ultra-Low NOx – SCAQMD 1146.2 Certified
- ♦ Cascade up to 4 Heaters – No External Sequencer Required



Patent Nos. 6,904,873 & 7,044,124;
other patents pending



Proudly Assembled in the USA

Heat Exchanger

- Headers
 - ☐ Brass – Standard
 - ☐ A-2 Glass-Lined Cast Iron
- ASME Inspected and HLW Stamped 160 PSIG Working Pressure
- National Board Approved
- Fin Tubing
 - ☐ Copper (Std.) 85%
 - ☐ Cupro-Nickel 83.1%
- ASME Steel Tube Sheet
- Silicone High-Temp. O-Rings
- ASME Pressure Relief Valve
 - ☐ 125 PSIG – Standard
 - ☐ _____ PSIG – Optional
- Temperature and Pressure Gauge
- Water Connections
 - ☐ Left – Standard
 - ☐ A-6 Right Hand

Controls

- 120V, 60Hz, 1Ø Power Supply
- 120/24V 60Hz Transformer
- Versa IC® Integrated Control
- 100% Shut-Off/Lockout
- Hot Surface Ignition
- Ignition Module
 - ☐ 3-Try – Standard
 - ☐ C-6 Single-Try
- Remote Flame Sensor
- Fixed High Limit, Manual Reset 240°F
- On/Off Power Switch
- Flow Switch
- Blocked Vent Pressure Switch
- Combustion Air Proving Switch
- Programmable Pump Time Delays
 - Heater
 - Recirculation
- Enable/Disable

Controls – cont.

- LCD Display; Status, Fault and Diagnostics
- Cold Water Protection Function
 - ☐ Optional CWR Variable-Speed Pump (See Cat. 5000.73)
- Modbus RTU BMS Port (Up to 115K Baud,
 - ☐ B-85 – BMS Gateway – BACnet MS/TP, BACnet IP, N2 Metasys or Modbus TCP
 - ☐ B-86 – BMS Gateway – LonWorks

Gas Train

- Manual Gas Shut-Off Valve(s)
- Gas Pressure Regulator
- Safety Shut-off Valve
- Firing Control Valve
- Firing Mode: 2-Stage (WH3)
- Natural/Propane Gas
- Gas Connection
 - ☐ Left - Standard
 - ☐ Z-1 Right
- Design Certified - ANSI Z21.10.3/CSA 4.3

Construction

- Indoor/Outdoor Construction
- Enclosed Front Controls
- Powder Coat Finish
- CSA Low Lead Certified (≤ .25% Lead)

Venting

- Vent Category I/III Cupro-Nickel
- Vent Category II/IV Copper
 - Vent Location
 - ☐ Top – Standard
 - ☐ D-14 Rear
 - Vent Termination Cap
 - ☐ D-11 Outdoor
 - ☐ D-15 Indoor, Horizontal
 - ☐ Indoor, Vertical (by others)
 - Combustion Air
 - ☐ D-17 In-Line Filter Kit (TruSeal only)
 - ☐ D-16 Air Intake Elbow

Venting – cont.

- Extractor – Optional
 - ☐ By others
 - ☐ Not required

Burner

- Ultra-Low NOx: Less Than 20 PPM

Options

- ☐ C-6 Ignition Module, Manual Rest
- ☐ D-21 TruSeal® Direct Vent
- ☐ F-10 Low Water Cut-Off, Remote Probe
- ☐ I-1 High Limit Control, Auto Reset
- ☐ P-___ Pump Bronze
 - _____ HP, 120V, 1Ø, 60Hz
 - Water Hardness _____ GPG
 - ☐ Mounted ☐ Loose
 - ☐ Front ☐ Rear
- ☐ P-___ Cold Water Run (See 1000.19)
- ☐ S-1 Low Gas Pressure Switch, Manual
- ☐ S-2 High Gas Pressure Switch, Manual
- ☐ X-1 SureRack® Kit
- ☐ X-2 SureRack Add-on

Additional Options or Accessories

☐ _____

Multi-Boiler Temperature Controllers

- ☐ TempTracker Mod+ Hybrid
 - ☐ B-36, 2-4 Heaters
 - ☐ B-37, 5-10 Heaters
 - ☐ B-38, 11-16 Heaters
 - ☐ B-39 EMS 4-20mA Remote Setpoint Interface Module
 - ☐ B-62 BACnet MS/TP Interface

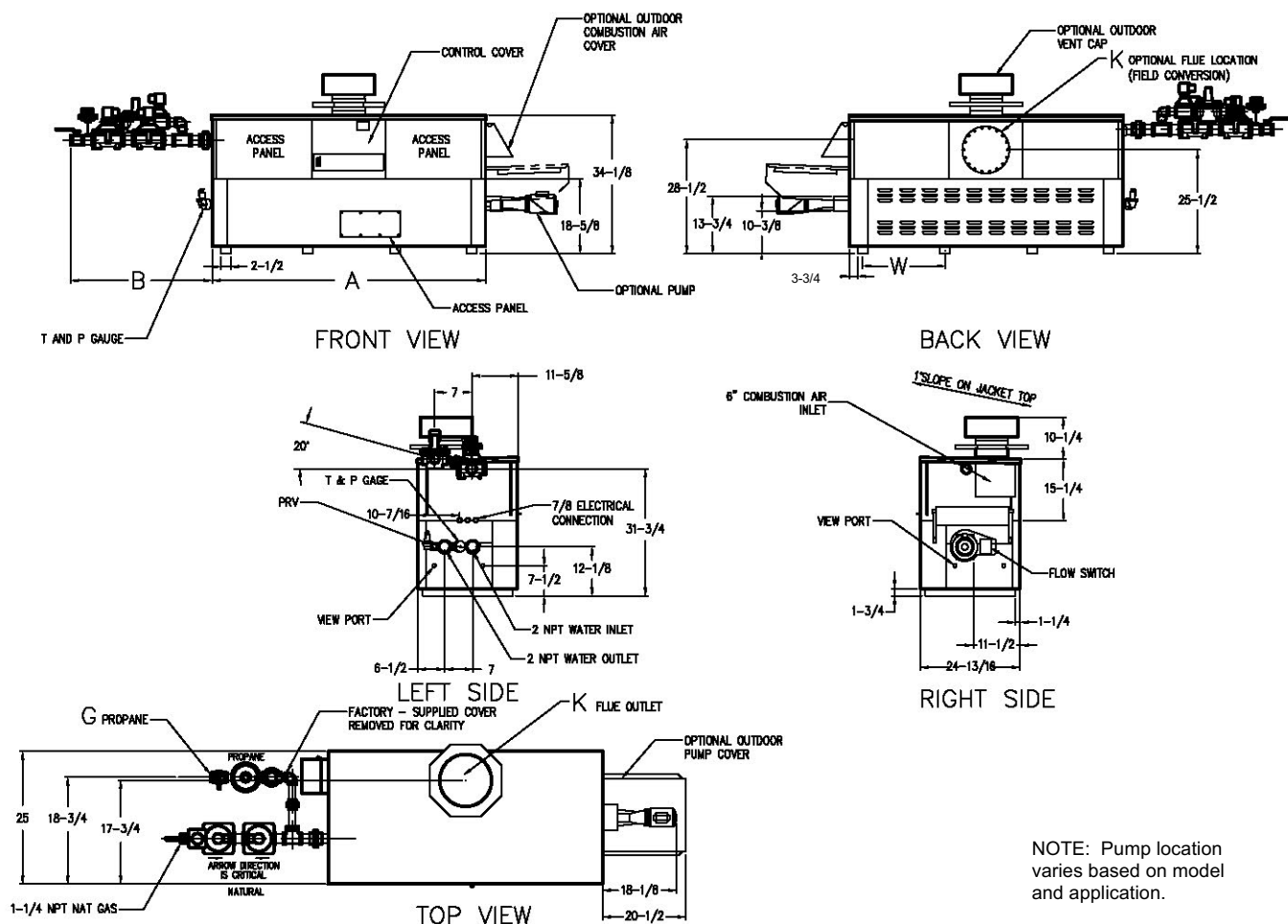


Raupak®
A Rheem® Company

Hi Delta Flex Gas – Type WH

Models 302CD-902CD

Model _____



NOTE: Pump location varies based on model and application.

	SPECIFICATIONS								Approx. Shipping Wt. (lbs.)
Model	MBTUH			A Width	G NPT	K	W	Amp. Draw*	
	Input	Output 85%	Output 83.1%						
☐ 302CD	300	255	249	36	3/4	6	14-3/4	6	380
☐ 402CD	399	339	332	43	3/4	6	18-1/2	6	445
☐ 502CD	500	425	416	50	1-1/4	6	22	6	545
☐ 652CD	650	553	540	60-1/2	1-1/4	8	27-1/4	6	590
☐ 752CD	750	638	623	67-1/2	1-1/4	8	30-3/4	6	675
☐ 902CD	900	765	748	78	1-1/4	8	36	7	740

* Without pump

- NOTES:
1. Dimensions are in inches.
 2. Rates shown are for natural or propane gas, and elevations up to 4,500 feet. For installation above 4,500 feet, please contact manufacturer.
 3. Required natural gas pressure is 7 – 10.5" WC. Required propane gas pressure is 11 - 13" WC.

Hi Delta Flex Gas – Type WH

Models 302CD-902CD

Model _____

Model	83.1% System Flow and Head Loss														
	3-4 Grains Per Gallon					5-15 Grains Per Gallon					16-25** Grains Per Gallon				
	ΔT (°F)	GPM	ΔP (FT WC)	MTS (IN)	SHL (FT WC)	ΔT (°F)	GPM	ΔP (FT WC)	MTS (IN)	SHL (FT WC)	ΔT (°F)	GPM	ΔP (FT WC)	MTS (IN)	SHL (FT WC)
☐ 302CD	13	40	2.0	2	5.1	10	50	3.0	2	7.7	6	90	9.8	2	23.7
☐ 402CD	17	40	2.0	2	5.1	13	52	3.4	2	8.4	7	90	10.0	2	23.8
☐ 502CD	21	40	2.1	2	5.2	16	52	3.5	2	8.6	9	90	10.4	2	24.2
☐ 652CD	27	40	2.2	2	5.3	21	52	3.7	2	8.7	12	90	10.8	2	24.7
☐ 752CD	28	45	3.0	2	6.8	23	55	4.4	2	9.9	14	90	11.3	2	25.1
☐ 902CD	28	54	4.3	2	9.7	21	73	7.8	2	17.2	17	90	11.7	2	25.5

SHL includes 50 ft of tubing each way (100 total)

** Must use optional cupro-nickel tubes. For hardness exceeding 25 GPG, must use water softener

Basis for minimum flow is 40 GPM or 30°F ΔT, 90 GPM for maximum flow

Model	85% System Flow and Head Loss														
	3-4 Grains Per Gallon					5-15 Grains Per Gallon					16-25** Grains Per Gallon				
	ΔT (°F)	GPM	ΔP (FT WC)	MTS (IN)	SHL (FT WC)	ΔT (°F)	GPM	ΔP (FT WC)	MTS (IN)	SHL (FT WC)	ΔT (°F)	GPM	ΔP (FT WC)	MTS (IN)	SHL (FT WC)
☐ 302CD	13	40	2.0	2	5.1	10	50	3.0	2	7.7	6	90	9.8	2	23.7
☐ 402CD	17	40	2.0	2	5.1	13	52	3.4	2	8.4	8	90	10.0	2	23.8
☐ 502CD	21	40	2.1	2	5.2	17	52	3.5	2	8.6	10	90	10.4	2	24.2
☐ 652CD	28	40	2.2	2	5.3	21	52	3.7	2	8.7	12	90	10.8	2	24.7
☐ 752Cd	29	45	3.0	2	6.8	23	55	4.4	2	9.9	14	90	11.3	2	25.1
☐ 902CD	29	54	4.3	2	9.7	21	73	7.8	2	17.2	17	90	11.7	2	25.5

SHL includes 50 ft of tubing each way (100 total)

** Must use optional cupro-nickel tubes. For hardness exceeding 25 GPG, must use water softener

Basis for minimum flow is 40 GPM or 30°F ΔT, 90 GPM for maximum flow

Model	83.1 % Recovery Rates								
	20°	30°	40°	50°	60°	70°	80°	90°	100°
☐ 302CD	1511	1007	755	604	504	432	378	336	302
☐ 402CD	2010	1340	1005	804	670	574	502	447	402
☐ 502CD	2518	1679	1259	1007	839	709	630	560	504
☐ 652CD	3274	2182	1637	1309	1091	935	818	727	655
☐ 752CD	3777	2518	1889	1511	1259	1079	944	839	755
☐ 902CD	4533	3022	2266	1813	1511	1295	1133	1007	907

Model	85 % Recovery Rates								
	20°	30°	40°	50°	60°	70°	80°	90°	100°
☐ 302CD	1545	1030	773	618	515	442	386	343	309
☐ 402CD	2055	1370	1028	822	685	587	514	457	411
☐ 502CD	2576	1717	1288	1030	859	736	644	572	515
☐ 652CD	3348	2232	1674	1339	1116	957	837	744	670
☐ 752CD	3864	2576	1932	1545	1288	1104	966	859	773
☐ 902CD	4636	3091	2318	1855	1545	1325	1159	1030	927

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